

CHAPTER 3: ELECTRIC CURRENT & DIRECT CURRENT CIRCUITS

At the end of this chapter, you'll be able to:

SUBTOPIC	LEARNING OUTCOMES
3.1	Electric current and direct current circuits a) Define and use electric current, $I = \frac{dQ}{dt}$ b) Use total charge, $Q = ne$ c) Define and use resistivity $\rho = \frac{RA}{l}$ d) State and use Ohm's Law in the circuit. e) Sketch $V - I$ graph for ohmic conductor. f) Use the effective resistance of various arrangements for Series: $R_{eff} = R_1 + R_2 + R_3 \cdots R_n$ Parallel: $\frac{1}{R_{eff}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \cdots + \frac{1}{R_n}$ <i>*limited to four resistors.</i> <i>*included using for Ohm's law in the circuit.</i>

3.1 Electric Current and Direct Current Circuits**Electric Current**

Electric current is defined as the rate of flow of charge.

$$I = \frac{dQ}{dt}$$

where dQ is the amount of charge that pass through a conductor at any location during the time interval, dt .

- It is base and scalar quantities. **The S.I. unit is ampere (A).**
- 1 ampere of current is defined to be **1 Coulomb** of charge flowing through the area in **1 second**.
- Can be measured using an ammeter connected in series in a circuit.

Total charge

Total charge, Q can be calculated using the equation:

$$Q = ne$$

where
 n is number of electrons
 e is charge of electron (1.6×10^{-19} C)

- **The S.I. unit is Coulomb (C).**

Example 1

The current in a bulb is 1.5 A. Calculate the amount of charge that flows through the bulb in 5.0 minutes and determine the number of electrons.

Answers:

$$I = 1.5 \text{ A}$$

$$dt = 5.0 \text{ minutes} = 300 \text{ s}$$

$$I = \frac{dQ}{dt} \rightarrow 1.5 = \frac{dQ}{300}$$

$$Q = 450 \text{ C}$$

$$Q = ne$$

$$450 = n(1.6 \times 10^{-19})$$

$$n = 2.81 \times 10^{21} \text{ electrons}$$

Resistance

Resistance is defined as the ratio of the potential difference V , across an electrical component to the current I , passing through it.

It is scalar quantity and the **S.I. unit is ohm (Ω).**

$$R = \frac{V}{I}$$

where
 V : Potential difference
 I : Electric current

Resistivity

Resistivity is defined as the resistance of a unit cross-sectional area per unit length of the material.

$$\rho = \frac{RA}{l}$$

where

ρ : Resistivity

R: Resistance

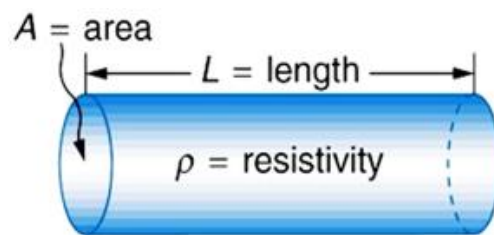
A: cross-sectional area

l : Length of the material

- Scalar quantity. **The S. I. unit is ohm meter (Ωm).**
- It is a measure of a material's ability to oppose the flow of an electric current.
- Resistivity **depends** on the **type of the material** and **temperature**.
- A good electric **conductors** have a very **low resistivities** and good **insulators** have very **high resistivities**.

Resistance of any wire or conductor

$$R = \frac{\rho l}{A}$$



The resistance of a wire or other conductor of uniform cross section is

- directly proportional to its length, l
- inversely proportional to its cross-sectional area, A
- directly proportional to the resistivity of the material of which the conductor is made, ρ

Ohm's Law

Ohm's law states that the **potential difference V**, across a metallic conductor is proportional to the **current I**, flowing through it if its physical conditions & temperature are constant. Mathematically, $V \propto I$.

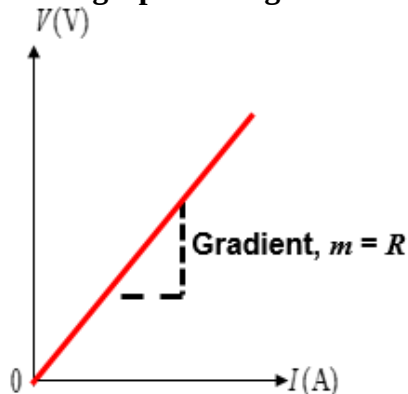
$$V = IR$$

where

V: Potential Difference, I: Current and R: Resistance of a conductor

Materials that obey Ohm's law (have constant resistance over a wide range of voltage) are called ohmic conductor.

Sketch graph of V against I for ohmic conductor



Example 2

A wire 4.0 m long and 6.0 mm in diameter has a resistance of 15 mΩ. A potential difference of 23.0 V is applied between the ends. Determine

- The current in the wire.
- The resistivity of the wire material.

Answers:

- $R = 15 \text{ m}\Omega = 15 \times 10^{-3} \Omega$
 $V = 23.0 \text{ V}$
 $V = IR \rightarrow 23 = (I)(15 \times 10^{-3})$
 $I = 1533 \text{ A}$
- $l = 4.0 \text{ m}$
 $d = 6.0 \text{ mm} = 6.0 \times 10^{-3} \text{ m}$
 $A = \frac{\pi d^2}{4} = \frac{\pi(6.0 \times 10^{-3})^2}{4}$
 $A = 2.83 \times 10^{-5} \text{ m}^2$
 $R = 15 \text{ m}\Omega = 15 \times 10^{-3} \Omega$
 $\rho = \frac{RA}{l} = \frac{(15 \times 10^{-3})(2.83 \times 10^{-5})}{4.0}$
 $\rho = 1.06 \times 10^{-7} \Omega\text{m}$

Example 3

The radius of a piece of 22-gauge tungsten wire is 0.321 mm and the resistivity of tungsten is $5.6 \times 10^{-8} \Omega\text{m}$.

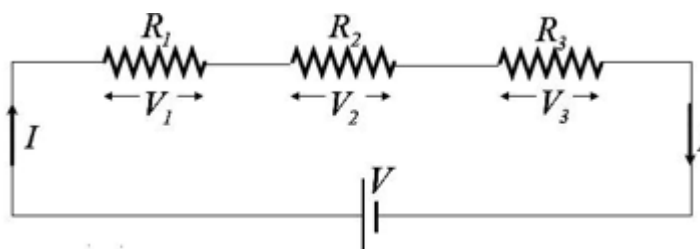
- Calculate the resistance per unit length of the wire.
- If a potential difference of 1.5 V is maintained across a tungsten wire of length 2.0 m, what is the current in the wire?

Answers:

- $r = 0.321 \text{ mm} = 0.321 \times 10^{-3} \text{ m}$
 $A = \pi r^2 = \pi(0.321 \times 10^{-3})^2 = 3.24 \times 10^{-7} \text{ m}^2$
 $\rho = 5.6 \times 10^{-8} \Omega\text{m}$
 $\frac{R}{l} = \frac{\rho}{A} = \frac{5.6 \times 10^{-8}}{3.24 \times 10^{-7}}$
 $\frac{R}{l} = 0.17 \Omega\text{m}^{-1}$
- $V = 1.5 \text{ V}$
 $l = 2.0 \text{ m} \rightarrow R = \frac{\rho}{A}l = (0.17)(2.0) = 0.34 \Omega$
 $V = IR \rightarrow I = \frac{V}{R} = \frac{1.5}{0.34}$
 $I = 4.41 \text{ A}$

Resistors in Series and Parallel

Resistors in series



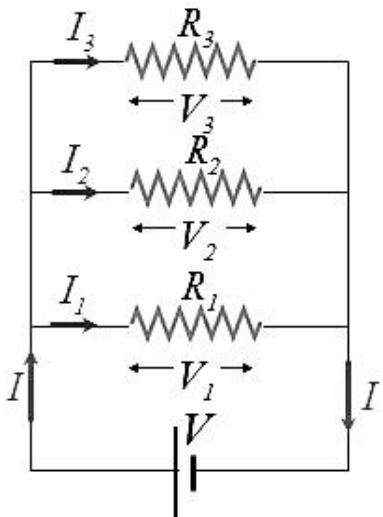
$$I = I_1 = I_2 = I_3$$

$$V = V_1 + V_2 + V_3$$

$$R_E = R_1 + R_2 + R_3$$

- The same current I passes through each resistor.
- Total voltage, V is shared by the resistors.
- Value of R_E is always greater than any individual resistance.

Resistors in Parallel



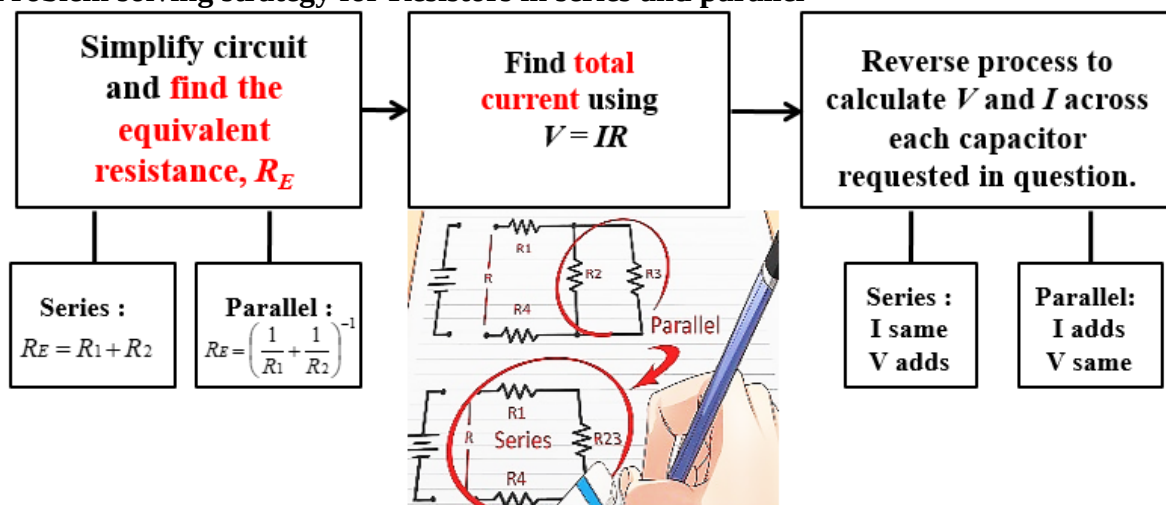
$$I = I_1 + I_2 + I_3$$

$$V = V_1 = V_2 = V_3$$

$$\frac{1}{R_E} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

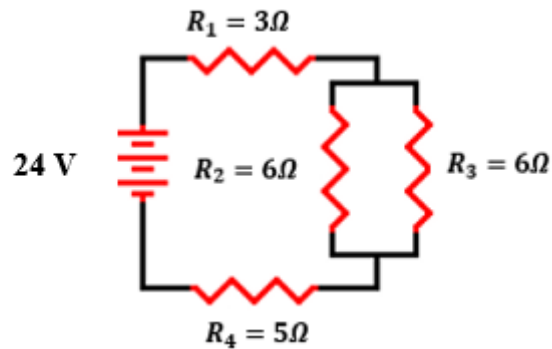
- Voltage, V same across each resistors.
- Total current, I splits into 3 separate path, I_1 , I_2 and I_3
- Value of R_E is always less than the smallest resistance in group.

Problem solving strategy for Resistors in series and parallel



Example 4

Figure shows a circuit consists of four resistors.

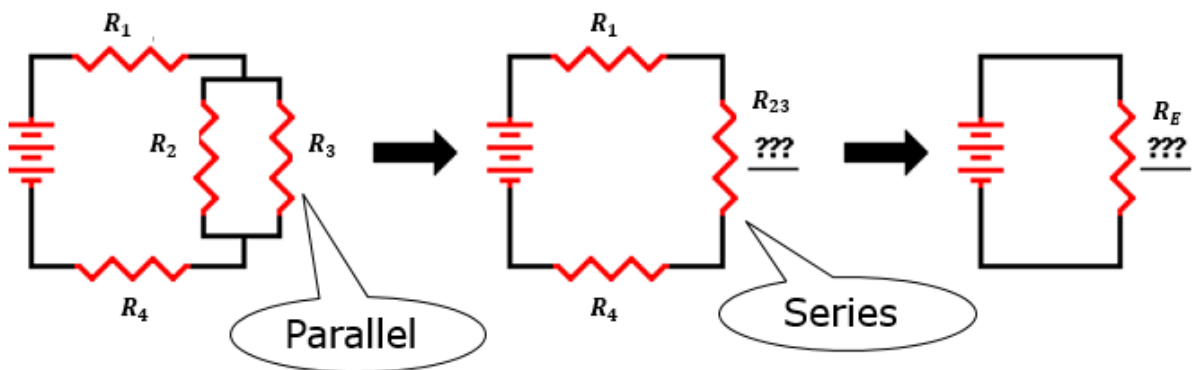


Calculate

- The equivalent resistance of the circuit.
- The total current in the circuit.
- The current flows through R_1 and R_4 .

Answers:

a)



R_2 and R_3 are in parallel

$$\frac{1}{R_{23}} = \frac{1}{R_2} + \frac{1}{R_3}$$

$$\frac{1}{R_{23}} = \frac{1}{6} + \frac{1}{6}$$

$$R_{23} = 3\Omega$$

R_1 , R_{23} and R_4 are in series

$$R_E = R_1 + R_{23} + R_4$$

$$R_E = 3 + 3 + 5$$

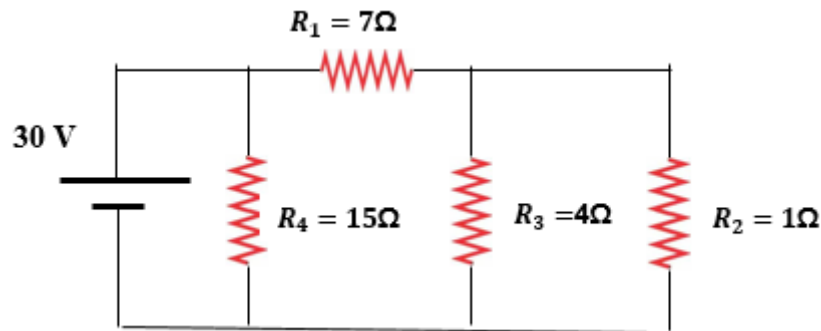
$$R_E = 11\Omega$$

b) $V = I_T R_E$
 $24 = (I_T)(11)$
 $I_T = 2.18\text{ A}$

c) R_1 , R_{23} and R_4 are in series so $I_T = I_1 = I_{23} = I_4 = 2.18\text{ A}$
 $I_1 = 2.18\text{ A}$
 $I_4 = 2.18\text{ A}$

Example 5

Figure shows a circuit consists of four resistors.

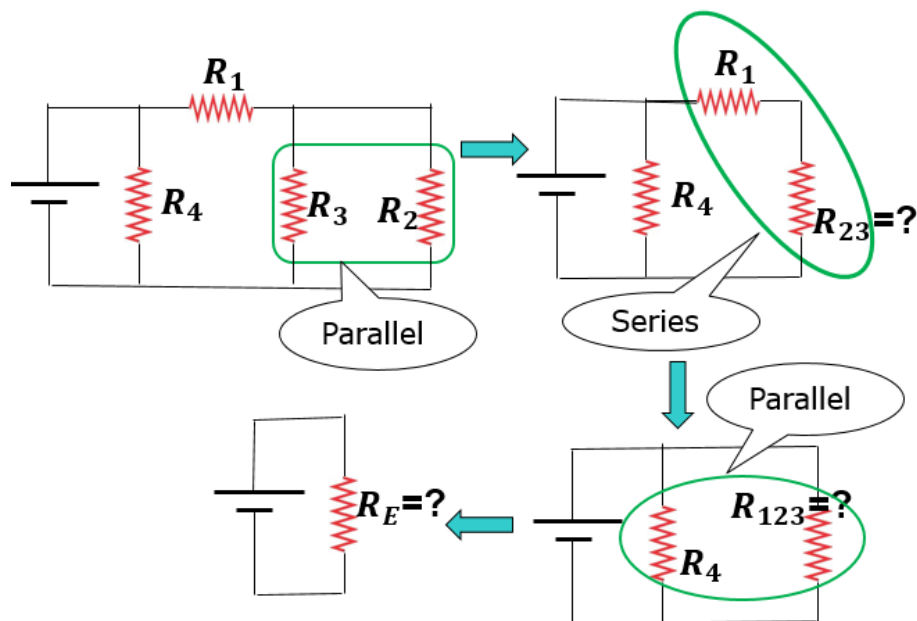


Calculate

- The equivalent resistance of the circuit.
- The total current in the circuit.
- The potential difference across R_4 .

Answers:

a)



R_2 and R_3 are in parallel

$$\frac{1}{R_{23}} = \frac{1}{R_2} + \frac{1}{R_3}$$

$$\frac{1}{R_{23}} = \frac{1}{1} + \frac{1}{4}$$

$$R_{23} = 0.8 \Omega$$

R_1 and R_{23} are in series

$$R_{123} = R_1 + R_{23}$$

$$R_{123} = 7 + 0.8$$

$$R_{123} = 7.8 \Omega$$

R_{123} and R_4 are in parallel

$$\frac{1}{R_E} = \frac{1}{R_{123}} + \frac{1}{R_4}$$

$$\frac{1}{R_E} = \frac{1}{7.8} + \frac{1}{15}$$

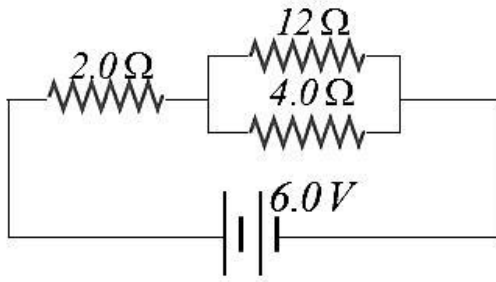
$$R_E = 5.13 \Omega$$

b) $V = I_T R_E$
 $30 = (I_T)(5.13)$
 $I_T = 5.85 \text{ A}$

c) R_{123} and R_4 are in parallel
 $V = V_{123} = V_4 = 30 \text{ V}$
 $V_4 = 30 \text{ V}$

Example 6

For the circuit shown below, calculate:



- The effective resistance of the circuit.
- The total current in the circuit.
- The value of current across each resistor.
- The potential difference across each resistor.

Answers:

- a) Let $R_1 = 2\ \Omega$; $R_2 = 12\ \Omega$; $R_3 = 4\ \Omega$

R_2 and R_3 are in parallel

$$\frac{1}{R_{23}} = \frac{1}{R_2} + \frac{1}{R_3}$$

$$\frac{1}{R_{23}} = \frac{1}{12} + \frac{1}{4}$$

$$R_{23} = 3\ \Omega$$

R_1 and R_{23} are in series

$$R_E = R_1 + R_{23}$$

$$R_E = 2 + 3$$

$$R_E = 5\ \Omega$$

- b) $V = I_T R_E$
 $6 = (I_T)(5)$
 $I_T = 1.2\ A$

- c) R_1 and R_{23} in series so $I_T = I_1 = I_{23} = 1.2\ A$
 R_2 and R_3 are in parallel so $V_{23} = V_2 = V_3$

$$V_{23} = I_{23} R_{23}$$

$$V_{23} = (1.2)(3)$$

$$V_{23} = 3.6\ V$$

$$V_{23} = V_2 = V_3 = 3.6\ V$$

$$V_2 = I_2 R_2$$

$$3.6 = (I_2)(12)$$

$$I_2 = 0.3\ A$$

$$V_3 = I_3 R_3$$

$$3.6 = (I_3)(4)$$

$$I_3 = 0.9\ A$$

$$\therefore I_1 = 1.2\ A; I_2 = 0.3\ A; I_3 = 0.9\ A$$

- d) R_2 and R_3 are in parallel so $V_{23} = V_2 = V_3 = 3.6\ V$

$$V_1 = I_1 R_1$$

$$V_1 = (1.2)(2)$$

$$V_1 = 2.4\ V$$

$$\therefore V_1 = 2.4\ V; V_2 = 3.6\ V; V_3 = 3.6\ V$$

Chapter 3 MCQs Practice:



SELECTED FORMULAE

1. $I = \frac{dQ}{dt}$

2. $Q = ne$

3. $\rho = \frac{RA}{l}$

4. $R_{eff} = R_1 + R_2 + R_3 \cdots R_n$

5. $\frac{1}{R_{eff}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \cdots + \frac{1}{R_n}$

6. $V = IR$

DIY Quiz

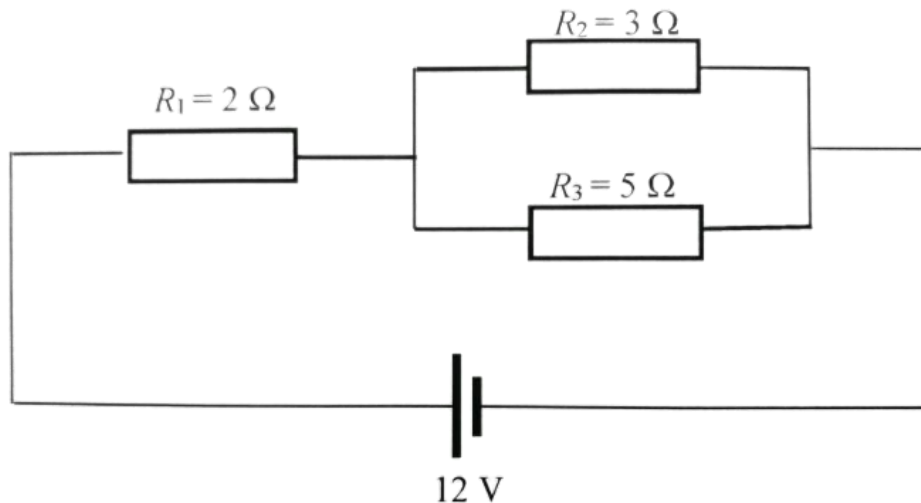
Instruction:

Answer all questions in 10 minutes.

No discussion & do not refer to any notes.

You can only refer to the formulae given below.

1. Figure shows a circuit with a 12 V battery connected with three resistors 2 Ω , 3 Ω and 5 Ω .



Calculate the

- Effective resistance.
- Voltage across 5 Ω resistor.

Answers:

- $R_E = 3.88 \Omega$
- $V_3 = 5.81 V$

PAST YEAR QUESTIONS

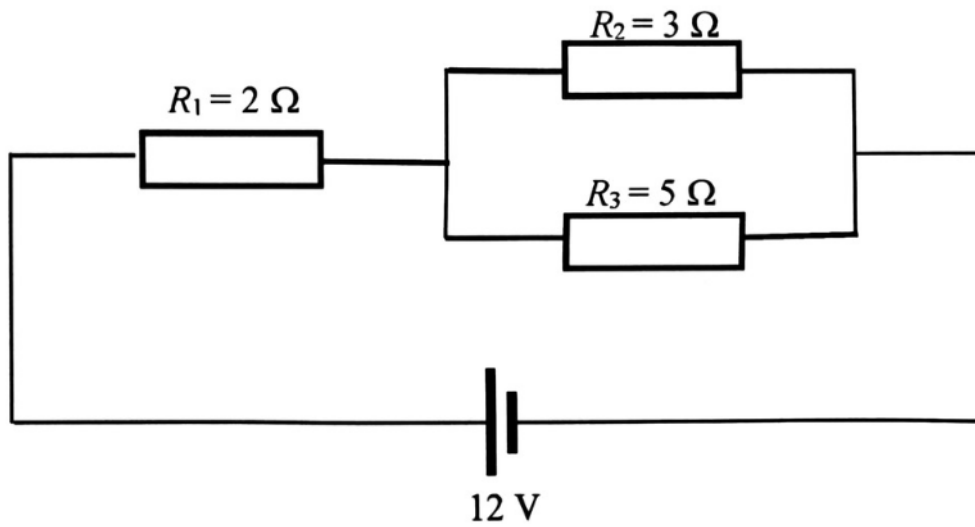


Figure above shows a circuit with a 12 V battery connected with three resistors 2 Ω , 3 Ω and 5 Ω . Calculate the

- a) Effective resistance.
- b) Voltage across 5 Ω resistor.

[7 marks]